

Prism BA on Cold Production Problems

Muell and Fuchsberg measurements against Caspar

Prism BA Research Project

Evidence report; frozen numbers, mixed-configurations labelled explicitly

13 September 2026

Abstract

This report isolates the available cold, production-style bundle-adjustment evidence for Prism and Caspar. The primary experiment uses one pre-registered Prism Config R on 22 Muell global-BA snapshots and five Fuchsberg dumps, with three repetitions per cell and solver clocks excluding input loading. Endpoint quality is effectively tied: all 22 Muell cells are ties and all five Fuchsberg cells lie within +0.077% of Caspar. Prism uses $31\times$ less total solver wall on Muell and $12.1\times$ less on Fuchsberg. The convergence-rate comparison is subtler: Caspar reaches a coarse 1% gap faster, while Prism is faster or tied near a 0.1% gap. A separate 8,788-camera, 16.75-million-observation measurement shows both measured Prism profiles beating Caspar FP32 and FP64 in endpoint and total wall. These are strong fastest-among-tested production-workload results, but they do not establish a universal fastest-solver claim.

1 What “cold problems” means here

The cold set is distinct from the repeatedly tuned BAL photo-tourism panel. It contains production global-BA states from two reconstruction workloads:

Family	Cells	Cameras	Observations	Repeats	Character
Muell	22	24–493	30K–314K	$N = 3$	Warm, well-conditioned in-mapper global-BA snapshots from a production scan.
Fuchsberg	5	526–8,788	1.08M–16.75M	$N = 3$	Large monocular conversion/global-BA dumps, including the largest production problem in this ledger.

The primary ledger fixes Config R before the sweep. Prism and Caspar use their own native solve clocks with loading excluded. Endpoint comparisons use a $\pm 0.01\%$ tie band where a win/tie count is reported; the Fuchsberg family statement instead reports the observed worst endpoint difference. Total-wall ratios compare completed configured budgets and therefore must be accompanied by time-to-gap measurements.

2 Primary cold-workload result: pre-registered Config R

Family	Quality result	Worst reported gap	Total wall ratio	1% gap ratio	0.1% gap ratio
Muell	22/22 ties	median -0.0000%	$31.0\times$	$1.08\times$	$2.62\times$
Fuchsberg	5/5 within tolerance	$+0.077\%$	$12.1\times$	$0.54\times$	$1.01\times$

All ratios in the last three columns are Caspar time divided by Prism time. A value greater than one favours Prism. Thus Prism has the lower configured total wall on both families. At a coarse 1% gap, Muell is roughly tied and Caspar is faster on Fuchsberg. At a tight 0.1% gap, Prism is $2.62\times$ faster on Muell and statistically tied in rate on Fuchsberg. This is why the report does not use Caspar’s long post-convergence crawl to claim a large time-to-equal-quality speedup.

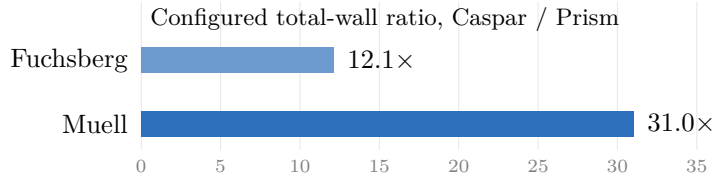


Figure 1: Total configured solver wall. These ratios include Caspar’s late crawl and are not, by themselves, convergence-rate ratios.

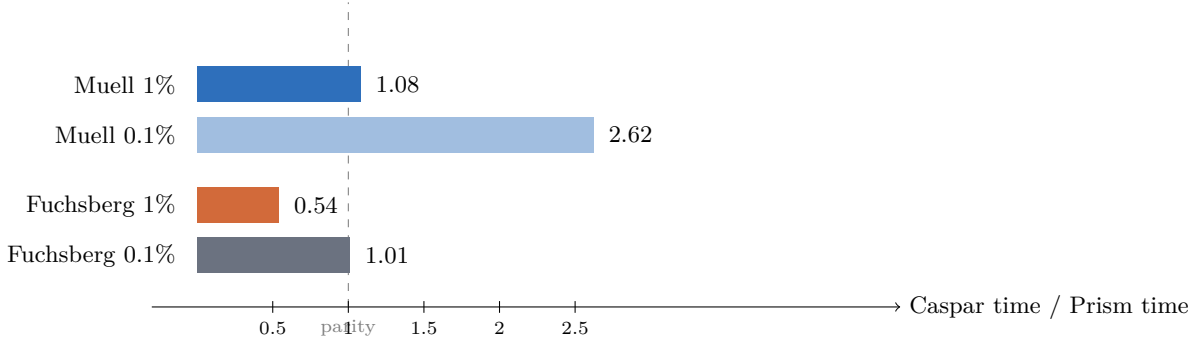


Figure 2: Descent-rate view. The apparent 12–31× total-wall advantage contracts sharply when both solvers are compared at the same quality gap.

3 Largest measured cold problem: Fuchsberg gba_234

The largest single problem has 8,788 cameras and 16.75 million observations. The table below is a separate measured configuration comparison retained from the production ledger; it is not Config R and is therefore not pooled into the preceding family tally.

Solver/configuration	Final cost	Native wall (s)	Caspar-FP64 wall / wall
Prism fast	3.0012×10^7	49.7	5.1×
Prism all-on	3.0017×10^7	27.5	9.2×
Caspar FP32	3.0037×10^7	95.0	2.7×
Caspar FP64	3.0019×10^7	252.0	1.0×

Both measured Prism profiles finish below both Caspar endpoints and use less wall. “All-on” is faster than “fast” in this row because these names identify frozen profiles rather than an ordering guaranteed on every problem.

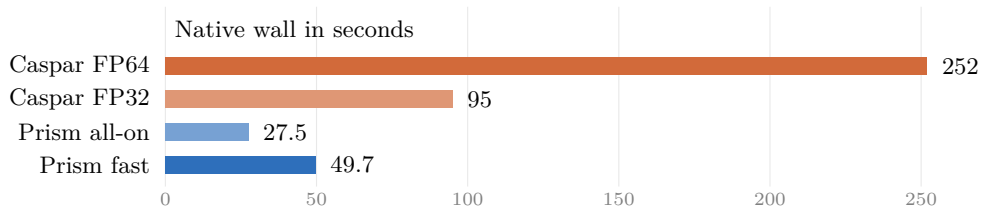
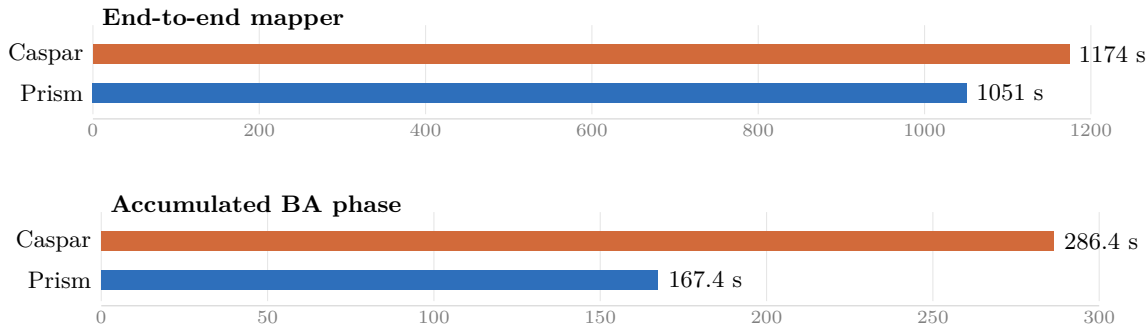


Figure 3: Configured wall on the 8,788-camera Fuchsberg problem.

4 End-to-end Muell mapper result

A separate N=3 interleaved incremental-mapping experiment measures the full application rather than isolated BA calls. Prism completes in 1,051 seconds versus 1,174 seconds for Caspar, a 10.5% end-to-end reduction with non-overlapping measured ranges. Both reconstruct all 493 registered images, with final reprojection error between 1.1099

and 1.1107 pixels. The isolated BA phase accounts for the difference: 167.4 seconds for Prism versus 286.4 seconds for Caspar, a 41.6% reduction over approximately 48–50 BA calls. Non-BA work agrees within 0.7%.



This is the strongest application-level cold result because it includes the consequences of solver choices on the surrounding mapper trajectory. It should be reported together with the equal map registration and reprojection statistics.

5 Independent Muell-gba146 target audit

The largest Muell snapshot has 493 cameras, 313,987 points and 2,118,671 observations. Its independently checked initial cost is 2,442,177.030208. A later Eta2/B6v7 run registered the target 1,946,488.7463 and reached it in all three repetitions at a median 4.0303 seconds (range 4.0263–4.0376), finishing at median cost 1,946,467.1524 after 16 outer iterations, zero rejects and 980 Schur products.

The separately frozen 90-second Caspar audit used the same input hash and independent FP64 rescoring:

Arm	Hits	Median audited cost	Median wall (s)	Median accepts / rejects
Prism Eta2/B6v7	3/3	1,946,467.15	4.030	16 / 0
Caspar FP32	0/3	1,961,895.56	90.006	960 / 682
Caspar FP64	0/3	1,962,982.12	90.145	568 / 0

This row shows a target that Caspar does not reach under its 90-second caps, so it supports a quality-and-budget statement rather than a finite time-to-target speed ratio. The FP32 median happens to be below the FP64 median by 0.055%; that difference is small relative to the distance to the registered target and should not be interpreted as a general precision ordering.

6 Claim boundary and missing evidence

The cold evidence supports the following wording:

On the measured Muell and Fuchsberg production workloads, Prism matches Caspar’s endpoint quality while reducing configured solver wall substantially. At identical quality gaps the advantage is workload- and tolerance-dependent: Prism is strongest near tight convergence on Muell, while Caspar reaches coarse accuracy faster on the largest Fuchsberg sequence. On the 8,788-camera measured configuration, Prism beats Caspar FP32 and FP64 in both endpoint and total wall.

It does not support “fastest BA solver” without qualification. The available local archive lacks the per-iteration Fuchsberg traces behind the five-cell aggregate, so this report includes summary figures rather than reconstructed convergence curves for those dumps. The raw Muell-gba146 Caspar rows are available and independently audited, but they use a different stopping budget from the Config-R family sweep. Future publication should restore the raw Fuchsberg traces, retain binary and input hashes per cell, and plot actual target crossings rather than infer them from total budgets.

Provenance

Primary Config-R family numbers and the total-wall/descent-rate correction come from `REPRODUCE.md`, Section 6c, in the synchronized research tree. The additional `gba_234` and end-to-end mapper rows come from the corrected product-workload section of `docs/results_2026_09.md`. The Muell-gba146 Caspar audit comes from `/workspace/prism-muell-caspar/summary.json`; the Eta2/B6v7 row comes from `research/eta2_wave5/b6v7-muell-summary.json`. No number in this report was inferred from a plot.